ENGINNERING CLINICS PROJECT

Review-3

GUIDE: Dr. Aruru Sai Kumar

Vehicle Accident Detection System

- Enhancing Road Safety Through Intelligent Systems

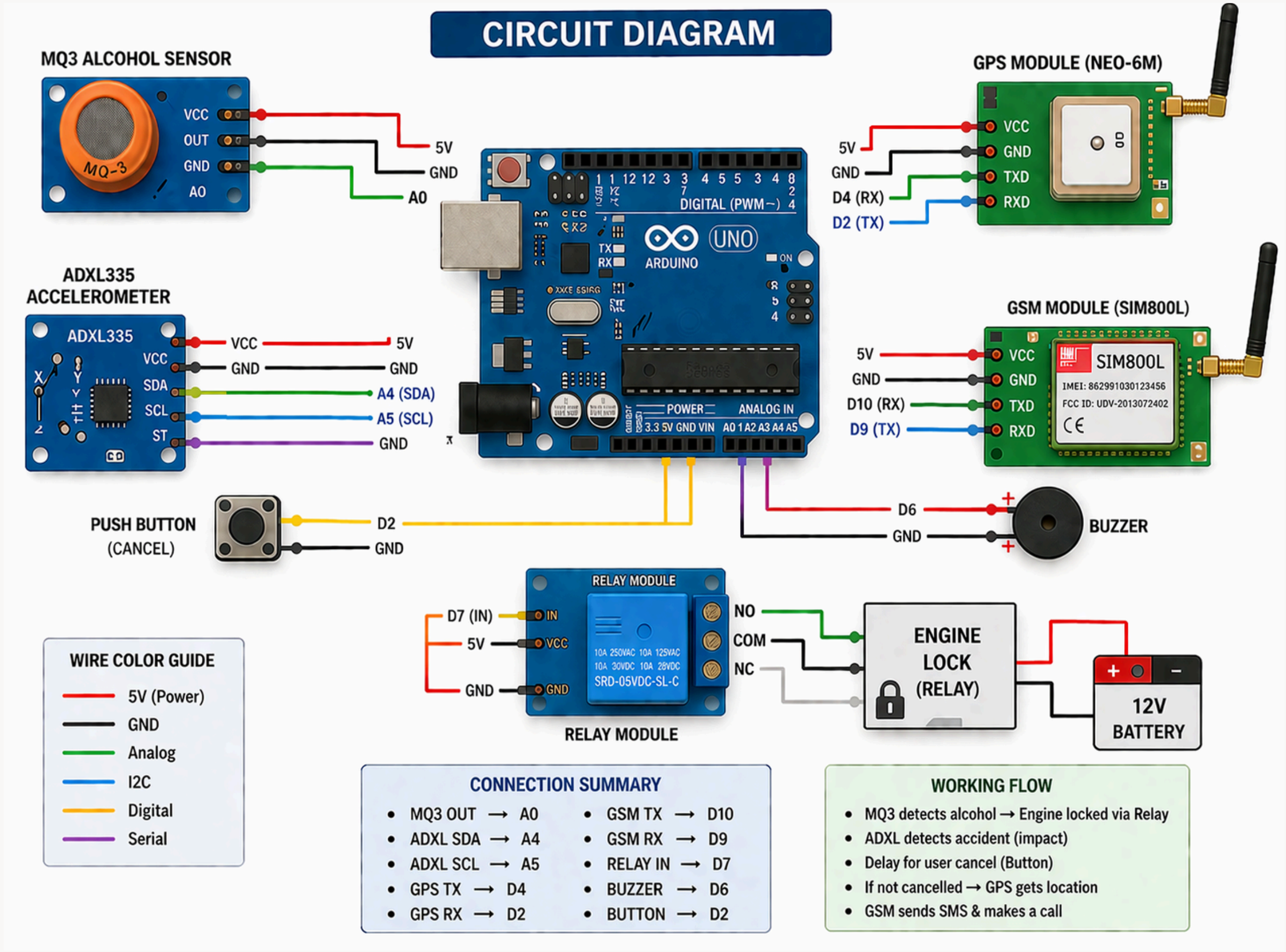
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SYSTEM WORKING

- System continuously senses driver condition (MQ3) and vehicle motion (ADXL)
- Alcohol level above threshold triggers engine locking via relay and buzzer alert
- Accelerometer detects sudden impact/tilt to identify accident conditions
- A delay mechanism is implemented to avoid false accident alerts
- Push button allows manual cancellation during the delay period
- GPS module acquires real-time latitude and longitude of the vehicle
- GSM module sends SMS and initiates a call with location details
- Final implementation ensures reliable GPS data acquisition and GSM communication

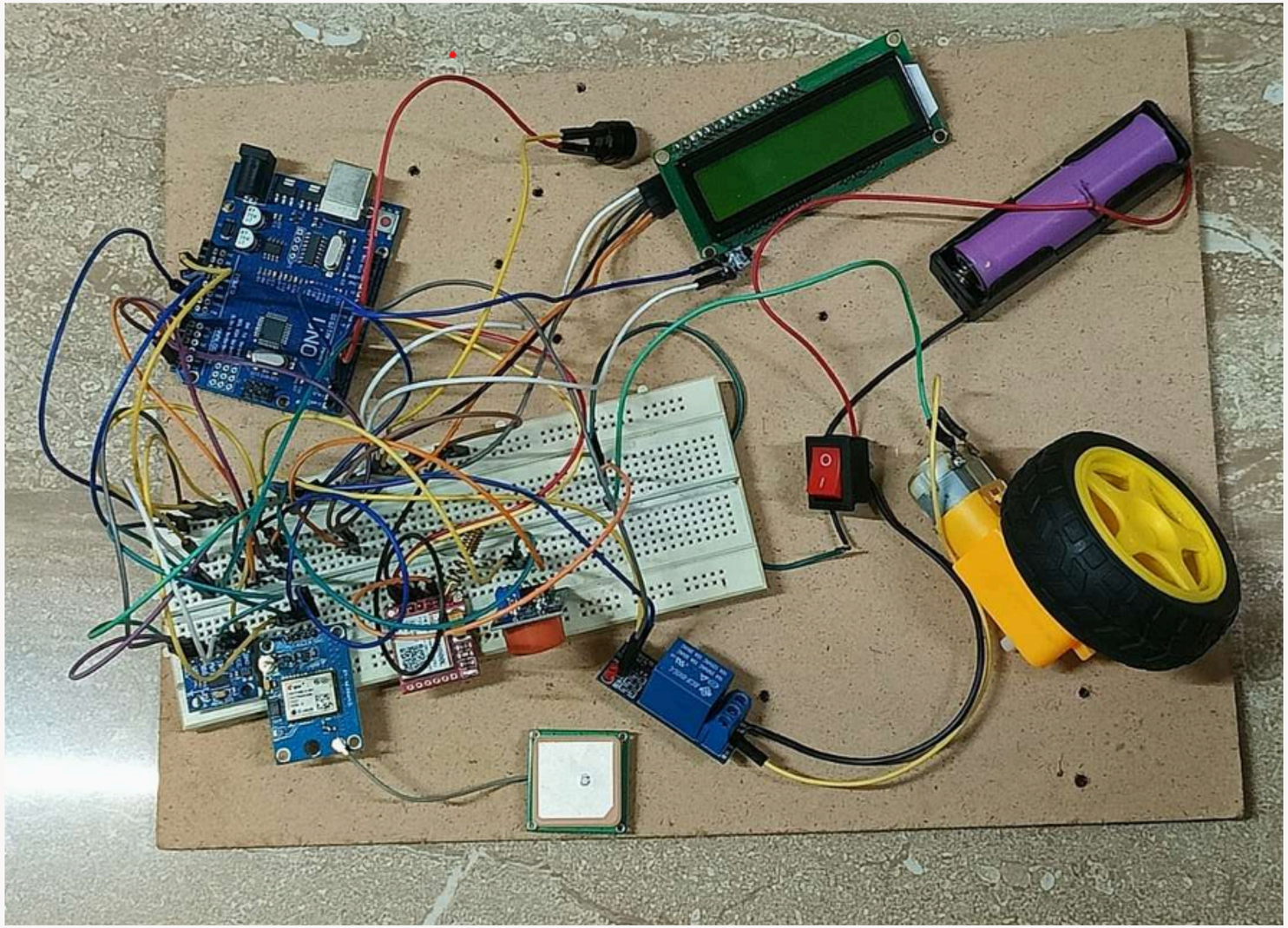
CIRCUIT DIAGRAM



EXPERIMENTAL SETUP & VERIFICATION

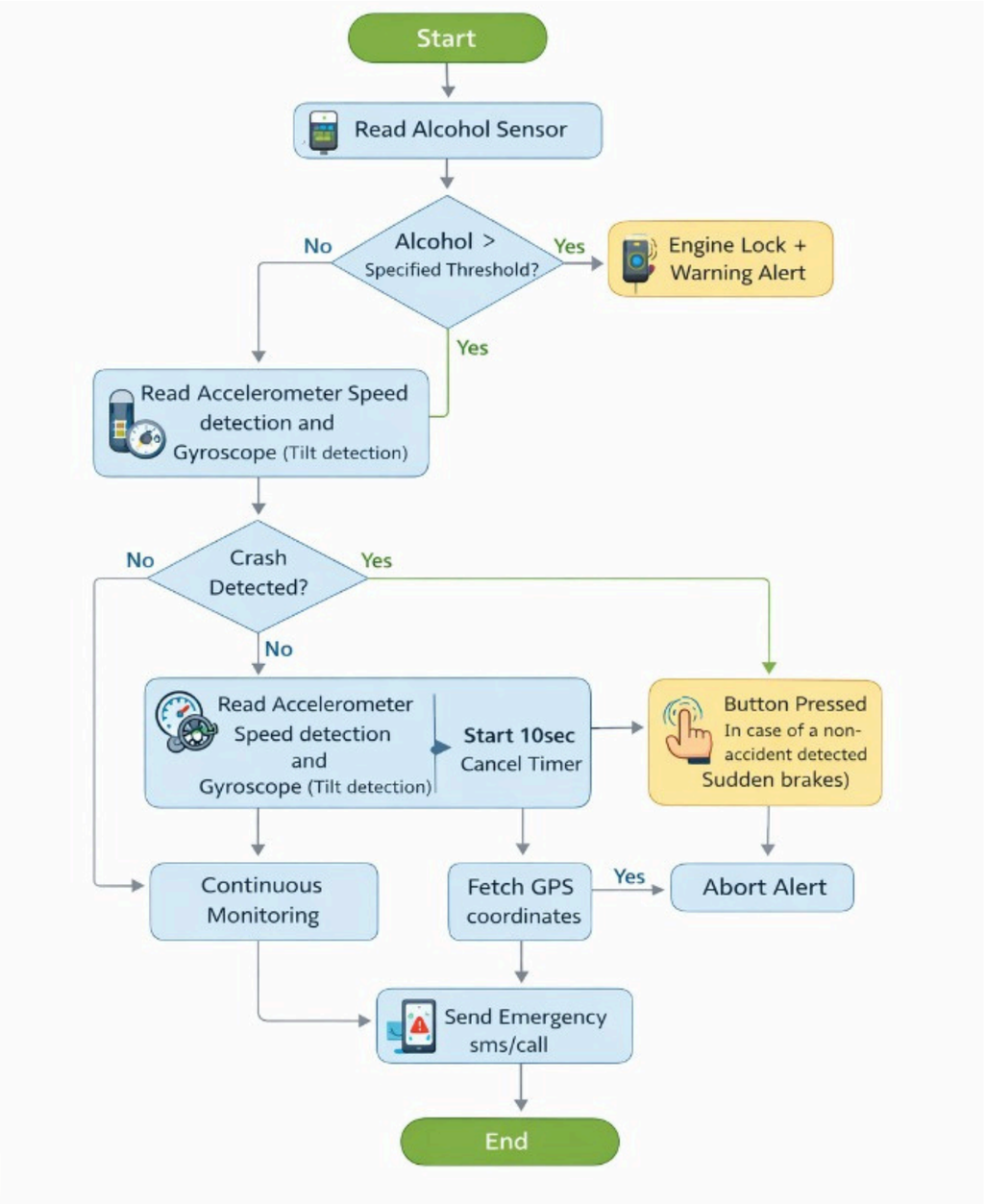
- All components are connected as per the circuit design using Arduino UNO
- System is powered using appropriate power supply for Arduino and GSM module
- Alcohol sensor is tested by exposing it to alcohol to verify engine locking
- Accelerometer is tested by applying sudden motion to simulate accident conditions
- GPS module is verified by checking real-time latitude and longitude output
- GSM module is tested by sending SMS and making calls to predefined numbers
- Push button is used to cancel false accident alerts during testing
- System performance is verified by integrating all modules and observing output

SYSTEM PROTOTYPE



WORKING PRINCIPLE

1. System starts and reads alcohol sensor
2. If alcohol > threshold → engine lock
3. Otherwise accelerometer monitors motion
4. Crash detected → 10 second timer starts
5. If button pressed → alert cancelled
6. Otherwise GPS location is fetched
7. GSM sends emergency message



RESULTS AND DISCUSSIONS

- Alcohol detection successfully prevents engine start when threshold is exceeded
- Accelerometer accurately detects sudden impacts indicating accident conditions
- Delay mechanism effectively reduces false accident alerts
- GPS module provides real-time location coordinates during emergencies
- GSM module successfully sends SMS and initiates calls with location details
- Integrated system performs reliably after improving GPS and GSM functionality
- Overall system demonstrates effective accident detection and alert mechanism



The MQ3 sensor measures alcohol level and compares it with a predefined threshold.
If the level exceeds the limit, the system locks the engine and displays status along with sensor readings on the LCD.



The system retrieves longitude coordinates from the GPS module to determine the exact position of the vehicle.
These coordinates are sent via GSM to provide precise location information in alert messages.



GPS Longitude Reading

The GPS module acquires real-time latitude coordinates of the vehicle's location using satellite signals.
This data is displayed on the LCD and used for accurate position tracking during emergencies.



Accident Detected

Timer for
Alert Cancellation

The accelerometer senses a sudden high impact indicating a possible accident condition.
A countdown timer is initiated, allowing the user to cancel the alert before automatic message transmission.



Alert Cancelled for False Accident Detection

The system detects a potential accident and starts a delay timer for confirmation.
If the user presses the push button within the delay, the alert is cancelled to avoid false notification.



Sending Alert when push button not pressed

When an accident is detected and no user response is given within the delay period, the system assumes it is a real emergency.
It then sends an alert message with GPS location using the GSM module.

TARGET ACHIEVEMENT:

- Successfully developed a working prototype of the system
- Achieved reliable alcohol detection and engine locking mechanism
- Implemented accurate accident detection using accelerometer
- Integrated GPS module for real-time location tracking
- Achieved successful GSM communication for sending alerts and calls
- Ensured overall system reliability after improvements in Review-III

REVIEW PROGRESS

Review-I:

- Identification of problem, project title, required components, and planning

Review-II:

- Development of working prototype with alcohol detection and accident detection modules

Review-III:

- Integration and improvement of GPS and GSM modules for reliable location tracking and alert system

LIMITATIONS OF THE PROJECT

- System depends on GSM network availability; alerts may fail in low-signal areas
- GPS accuracy may vary in tunnels, dense buildings, or remote locations
- Alcohol sensor (MQ3) may give false readings due to environmental factors
- No real-time monitoring of driver health condition (heart rate, fatigue, etc.)
- System is prototype-level, not fully optimized for real vehicle integration
- Power consumption and hardware size are not optimized for commercial use

FUTURE SCOPE

- Integration of steering-mounted pulse sensor to monitor driver heart rate
- Addition of BP (blood pressure) monitoring system for detecting health risks
- Use of eye-blink sensor / IR sensor to detect driver drowsiness or sleep
- Automatic engine control/stop if abnormal health condition or sleep is detected
- Implementation of camera-based AI system for driver behavior monitoring
- Use of advanced sensors (gyroscope + GPS fusion) for more accurate accident detection
- Mobile app integration for instant alerts and live tracking

CONCLUSION

- The system successfully detects accidents and prevents drunk driving
- Real-time location tracking and emergency alert system is achieved
- Reliability of GPS and GSM modules is improved in final implementation
- Overall, the system enhances road safety and reduces emergency response time

THANK YOU